Smart Router

Track 1 — General-Purpose AI Agent

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AMD Developer Hackathon — ACT II

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`ghcr.io/wilfred-dore/track1-smart-router:latest`

The problem: accuracy is a gate, tokens are the score

Enterprise reality

- Not every task deserves a premium model
- Host cheap local models in-house, call the paid API only when it matters
- Track 1 scores exactly that: pass the 80% accuracy gate, then rank by **fewest Fireworks tokens**

Consequences for design

- Local inference = 0 tokens ⇒ use it aggressively
- Every escalation must be *deliberate*
- Extra accuracy above the gate is wasted tokens

8 task categories: factual QA · math · sentiment · summarization · NER · code debugging · logic · code generation

Architecture: a single-pass cascade, never a loop

1. Deterministic solvers — arithmetic parser, sentiment lexicon, NER gazetteers, capital lookup — **0 tokens**

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2. Local LLM — Qwen2.5-3B, 4-bit GGUF, llama.cpp, CPU-only, bundled in the image — **0 tokens**

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3. Confidence gate — per-category heuristics; unreliable categories (math, logic) skip straight to 4 — **0 tokens**

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4. Fireworks escalation — model resolved from `ALLOWED_MODELS` at runtime, tight per-category output contract — **billed**

Engineering for the grading environment

- **Everything config-driven** — gate thresholds, category→model mapping, token budgets, escalation policy. No hardcoded answers, no caching across runs.
- **Reasoning-model safety** — inline <think> blocks stripped; if thinking consumes the whole token budget, the answer is recovered from the reasoning field instead of returning empty.
- **Graceful degradation** — a broken local runtime falls back to full escalation; the container never crashes, every `task_id` always gets an answer.
- **Mock mode** — the full pipeline runs and counts tokens without any API key; `auto` mode switches to live when the harness injects credentials.
- **Calibrated by sweep, not by hand** — routing policies compared on a 19-task local eval (8 official practice tasks + 11 self-authored variants).

Results — measured in CI under the exact grading limits

Metric	Measured	Constraint
Runtime, 19 tasks (4 GB RAM, 2 vCPU, amd64)	90 s	600 s
Compressed image size	1.97 GB	10 GB
Estimated Fireworks tokens / full run	∼900	leader: 4,268
Local eval accuracy (escalations assumed)	19/19	gate: 16/19
Output schema & exit code	valid	required

- CI pipeline: native amd64 build → smoke test at grading limits → size check → publish
- The smoke test already caught one real bug (musl-linked wheel) before any submission slot was spent

Takeaways

- **Zero-cost first**: solvers + a bundled 3B handle most of the load; the paid API sees only the hard residue
- **Route by reliability, not by difficulty**: categories where a small model fails *silently* (math, logic, multi-part factual) escalate outright
- **Plumbing wins hackathons**: most failed submissions break on infrastructure — CI reproduces the grading environment exactly

Submission

`ghcr.io/wilfred-dore/track1-smart-router:latest`

public · linux/amd64 · weights bundled · no secrets inside